

Isolated MIDI input / 3.3 V UART

M1 | INDEPENDENT MODULE | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J202 / MIDI IN
1: DIN4 IN 2: DIN5 IN

J101 / PWR IN / UART OUT
1: 5V IN 2: 3V3 IN
3: GND 0V 4: UART OUT

J203 / CASE BOND
1: CASE 2: GND 0V

U201.3: NC -- LEAVE OPEN

U201
onsemi H11L1M

J101
POWER INPUT + UART OUTPUT TO HOST

+5V 1
+3V3 2
GND 3
4

J203
METAL CASE BOND + GROUND

CHASSIS 1
GND 2

DIN panel socket: contact 4 -> J202.1; contact 5 -> J202.2. Pins 1/2/3 and shell unconnected.
Insulate DIN shell. J203.1 bonds the enclosure; R101 returns ESD to local GND.
Optocoupler LED current -> UART low. The output pull-up is 3.3 V, never 5 V.
Functional ground-loop isolation; transient clamps limit common-mode range. Not a safety isolator.

+3V3 +5V CHASSIS GND

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_MIDI.kicad_sch

Title: Isolated MIDI input / 3.3 V UART

Size: A3 Date: 2026-10-02
KiCad E.D.A. 9.0.2

Rev: M1
Id: 1/1

Passive expression pedal / I2C ADC

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J302 / PASSIVE PEDAL
1: TIP I/O 2: RING I/O
3: GND 0V

JP301 / PEDAL JUMPER
1: TIP 2: RING
3: PEDAL IN 4: EXC OUT
5: RING 6: TIP

JP303 / DATA PULLUP
1: PULLUP 2: DATA I/O

J101 / PWR IN / I2C
1: 3V3 IN 2: GND 0V
3: DATA I/O 4: CLOCK IN

JP302 / ADDR JUMPER
1: LOW GND 2: SELECT
3: HIGH 3V3

JP304 / CLOCK PULLUP
1: PULLUP 2: CLOCK IN

U301.2: UNUSED ALERT - LEAVE OPEN

J302
PASSIVE PEDAL INPUT / EXCITATION
OUTPUT

JP301
JUMPER ONLY: PEDAL POLARITY

R301

+3V3_A

1k / 1%

EXP_EXC

D303

BAT54H.115

+3V3_A

GND

D304

BAT54H.115

R302

10k / 1%

R303

1M / 1%

GND

C301

100n / 50V X7R

GND

R304

10k / 1%

EXP_EXC

D305

BAT54H.115

+3V3_A

C302

100n / 50V X7R

GND

D306

BAT54H.115

+3V3_A

U301

ADS1115IDGSR

ALERT/RDY

AIN0

AIN1

AIN2

AIN3

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

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GND

R308

2.2k / 1%

+3V3_A

JP303

JUMPER ONLY: SDA PULLUP

SDA

R309

2.2k / 1%

+3V3_A

JP304

JUMPER ONLY: SCL PULLUP

SCL

R305

33R / 1%

SCL

R306

33R / 1%

SDA

J101

3V3 POWER INPUT + I2C TO HOST

+3V3

GND

GND

GND

JP302

JUMPER ONLY: ADC ADDRESS

GND

GND

GND

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Engineering prototype – bench validation required

Ardor

Sheet: /

File: Ardor_Expression.kicad_sch

Title: Passive expression pedal / I2C ADC

Size: A3

Date: 2026-10-02

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Id: 1/1

+3V3

+3V3_A

CHASSIS

GND

ARDOR / INDEPENDENT FUNCTION MODULE

Stereo buffer / mono average / AC outputs

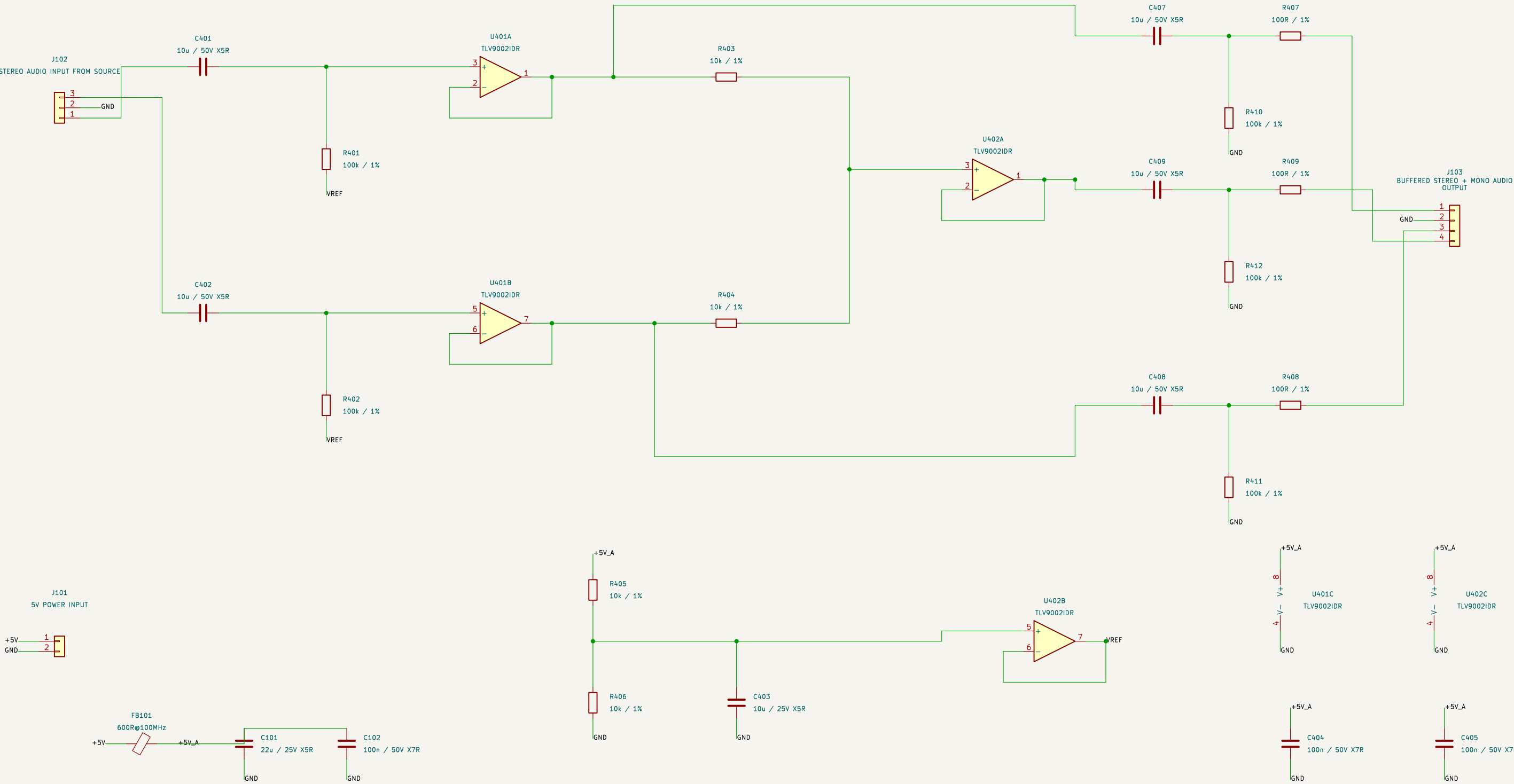
M1 | INDEPENDENT MODULE | SEE PIN GUIDE AT RIGHT

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J101 / POWER IN
1: 5V IN 2: GND 0V

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

J103 / AUDIO OUT
1: LEFT OUT 2: GND 0V
3: RIGHT OUT 4: MONO OUT



Line-level input ≤ 1 Vrms/channel. Output loads $\geq 10k$. Mono = (L + R)/2, unity buffers.
All three outputs AC coupled; no 2.5 V bias leaves this board. For mono input, feed both channels
or accept -6 dB when using one channel. Harness connectors are internal; add jack-entry ESD protection
for exposed ports. This board is neither an instrument preamp nor an ADC, headphone or power amplifier.

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Mixer.kicad_sch

Title: Stereo buffer / mono average / AC outputs

Size: A3 Date: 2026-10-02

Rev: M1

KiCad E.D.A. 9.0.2

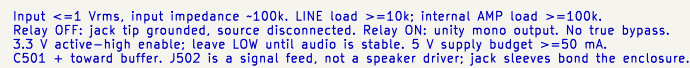
Id: 1/1

Mono line output / relay mute / internal amp feed

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J102 / AUDIO IN
1: AUDIO IN 2: GND 0V

J502 / AMP AUDIO OUT
1: AUDIO OUT 2: GND 0V



Id: 1/1

Stereo headphone amplifier / optional module

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

The schematic diagram illustrates the internal circuitry of a TPA6132A2RTER audio amplifier. The power supply section at the bottom left features a 5V input (J101) connected to a 600R resistor (FB101) and decoupling capacitors C101 (22uF) and C102 (100nF). The main amplifier U601 (TPA6132A2RTER) is powered by +5V_A, HPVDD, and HPVSS. It has two stereo audio inputs (J102) and two stereo headphone outputs (J602). The outputs are connected to a common chassis ground through resistors R601, R602, R603, and R604. The circuit also includes two diodes D601 and D602 for protection.

+5V +5V_A CHASSIS GND

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